REINS | first-cut digital core floorplan

UNPROTECTED ROW EXPERIMENT

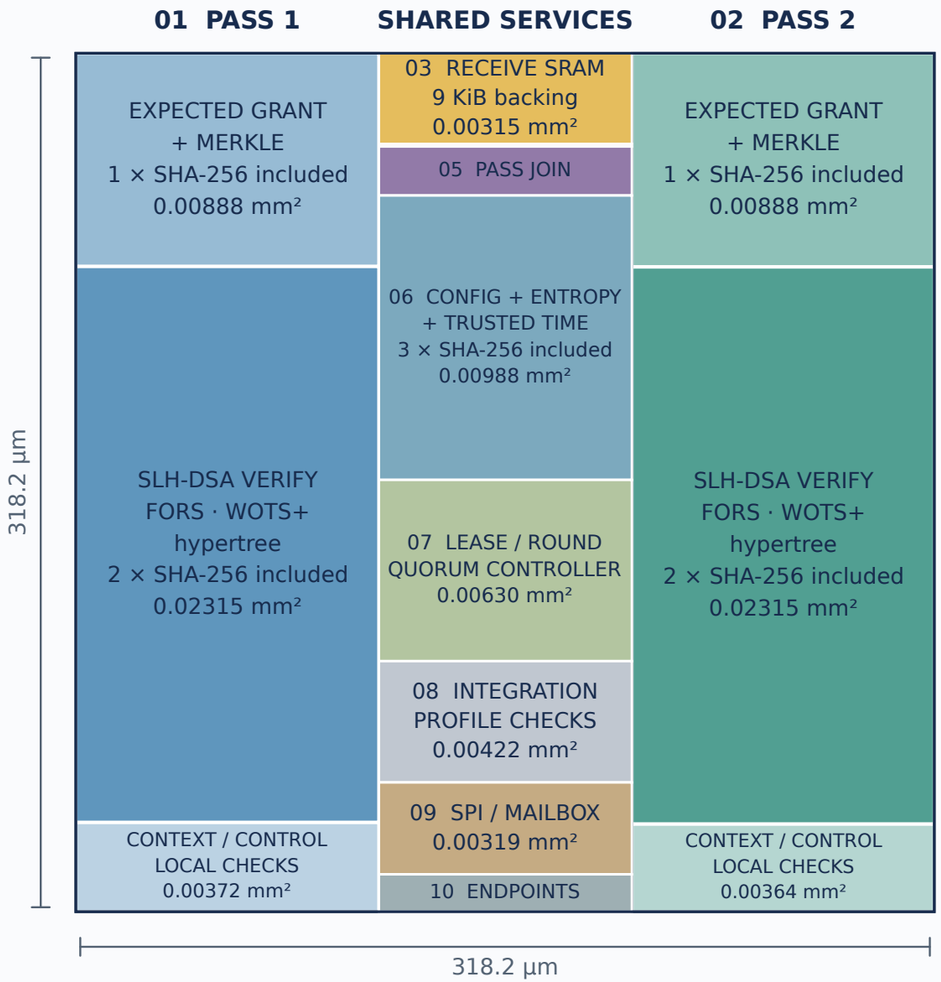
N3 planning scenario · area-proportional regions · not a placed or routed chip

2,725 kGE
Nangate45 reference logic

9 KiB SRAM
9 KiB payload; backing bits shown

135,954 state bits
Registers included in logic kGE

0.101 mm²
≈ 318 × 318 µm core



Estimated region areas

Logic regions include placement/routing allowance; SRAM is separate.

REGION	kGE	mm²
01 Verification pass 1	992.94	0.03575
02 Verification pass 2	990.84	0.03567
03 Receive SRAM macro	—	0.00315
04 RX checks / ownership	2.52	0.00009
05 Protected pass join	47.28	0.00170
06 Config / entropy / time	274.58	0.00988
07 Lease / round / quorum	174.98	0.00630
08 Integration / profile	117.09	0.00422
09 SPI / mailbox	88.72	0.00319
10 Endpoint guards / demo	36.11	0.00130
TOTAL DIGITAL CORE	**2,725.07**	**0.10125**

Planning envelope: 0.056–0.165 mm²

Density and macro assumptions; not a foundry estimate or signoff bound.

Pass regions are illustrative. Physical independence is not established.

Planning point: 0.018 µm²/GE · 0.0225 µm²/SRAM bit · logic ×1.25 / 62.5% utilization

SRAM macro = bitcell floor ×1.90. No test RTL included; three representative endpoint workloads are included. Nine SHA cores total.

NOT SIZED: OTP (272 bytes), physical entropy sources, timing references/sensors, pads/ESD, power/DFT, and the protected application processor.

Endpoint subtotal is an accounting region; actual guards belong at their physical isolation endpoints. This is a core sketch, not a die outline.

Source: measured production RTL at edb48ec · Yosys 0.52 / ABC-fast / Nangate45 · current non-comment RTL checked against that baseline

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